

Task 4: Optimal Switch Results

CS24BTECH11036 CS24BTECH11037

Assumptions

Binary Search is performed. Assuming plan cost is monotonous

Bounds of Binary search in case of Delayed switch is the date of previous switch and the date of the current switch

Bounds of Binary search in case of Early switch is the date of current switch and the date of the next switch - 1

Choosing the right most bound for an Early switch is questionable. because the right bound being the max_date and the next switch both can be incorrect. The case where the next switch is also early is a counter for next switch to be used as a bound.

The case where there does not exist an optimal switch till the next switch but exists after is a counter to the case where max_date is used as a bound.

extremes are replaced with min_date and max_date.

Query 1 Optimal Switches

Switch 1 (EARLY)

Original Switch: 1992-01-07 → 1992-01-08

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-01-26)	3495.67 ms	9303.47 ms
q'_j (1992-01-27)	3713.80 ms	10104.20 ms

Plan **Pi** is faster throughout the interval from the current date to next switch implying the optimal execution wouldn't perform any switches, Optimal date vouches for this as it is same as the next switch's date-1.

Switch 2 (DELAYED)

Original Switch: 1992-01-27 → 1992-01-28

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-01-08)	1370.07 ms	290.72 ms
q'_j (1992-01-09)	1559.17 ms	359.17 ms

Plan **Pj** is faster throughout the interval from current date to previous switch implying the optimal execution would have switched rightaway, Optimal date vouches for this as it is same as the previous switch's date.

Switch 3 (DELAYED)

Original Switch: 1992-03-29 $\rightarrow$ 1992-03-30

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-03-02)	9422.00 ms	9339.57 ms
q'_j (1992-03-03)	9662.28 ms	9281.83 ms

Plan **Pj** is faster throughout the interval from current date to previous switch implying the optimal execution would have switched rightaway, Optimal date vouches for this as it is same as the previous switch's date.

Query 2 Optimal Switches

Switch 2 (DELAYED)

Original Switch: 1992-02-18 $\rightarrow$ 1992-02-19

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-01-14)	5815.20 ms	5825.69 ms
q'_j (1992-01-15)	6062.06 ms	6124.84 ms

Plan **Pi** is faster throughout the interval from current date to the previous switch. which is impossible considering the plan cost is monotonous and considering the plan costs being almost same , The only conclusion that can be made is there was a spike in **Pj**'s cost in this interval causing issue.

Switch 3 (EARLY)

Original Switch: 1992-03-17 $\rightarrow$ 1992-03-18

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-05-16)	12155.90 ms	13635.69 ms
q'_j (1992-05-17)	12182.58 ms	13548.32 ms

Plan **Pi** is faster throughout the interval from the current date to next switch implying the optimal execution wouldn't perform any switches, Optimal date vouches for this as it is same as the next switch's date-1.

Switch 4 (DELAYED)

Original Switch: 1992-05-17 $\rightarrow$ 1992-05-18

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-03-18)	15943.09 ms	11953.03 ms
q'_j (1992-03-19)	15900.75 ms	11966.02 ms

Plan **Pj** is faster throughout the interval from current date to previous switch implying the optimal execution would have switched rightaway, Optimal date vouches for this as it is same as the previous switch's date.

Switch 5 (EARLY)

Original Switch: 1992-05-18 $\rightarrow$ 1992-05-19

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1997-12-23)	13458.81 ms	14540.90 ms
q'_j (1997-12-24)	13638.26 ms	14397.85 ms

Plan **Pi** is faster throughout the interval from the current date to next switch implying the optimal execution wouldn't perform any switches, Optimal date vouches for this as it is same as the next switch's date-1.

Switch 6 (DELAYED)

Original Switch: 1997-12-24 $\rightarrow$ 1997-12-25

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1997-06-25)	14230.71 ms	14379.75 ms
q'_j (1997-06-26)	14160.82 ms	14570.14 ms

Plan **Pi** is faster throughout the interval from current date to the previous switch. which is impossible considering the plan cost is monotonous and considering the plan costs being almost same , The only conclusion that can be made is there was a spike in **Pj**'s cost in this interval causing issue.

Query 3 Optimal Switches

Switch 1 (DELAYED)

Original Switch: 1992-01-03 → 1992-01-04

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-01-02)	1011.43 ms	772.38 ms
q'_j (1992-01-03)	1219.01 ms	1366.33 ms

Consider the data **Pj** is faster at **qi** and slower at **qj** considering the switch is delayed **Pj** is faster at the current data, this clearly shows that the plan cost is not monotonous and that is the reason why the algorithm couldnt find an optimal switch.

Switch 2 (DELAYED)

Original Switch: 1992-01-06 → 1992-01-07

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-01-05)	1999.65 ms	2077.26 ms
q'_j (1992-01-06)	2356.27 ms	2473.42 ms

Plan **Pi** is faster throughout the interval from current date to the previous switch. which is impossible considering the plan cost is monotonous The only conclusion that can be made is there was a spike in **Pj**'s cost in this interval causing issue.

Switch 3 (EARLY)

Original Switch: 1992-01-17 → 1992-01-18

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-02-15)	7812.58 ms	15994.83 ms
q'_j (1992-02-16)	7896.73 ms	15988.56 ms

Plan **Pi** is faster throughout the interval from the current date to next switch implying the optimal execution wouldn't perform any switches, Optimal date vouches for this as it is same as the next switch's date-1.

Switch 4 (DELAYED)

Original Switch: 1992-02-16 → 1992-02-17

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-01-18)	13588.39 ms	12204.93 ms
q'_j (1992-01-19)	13705.77 ms	12242.23 ms

Plan **Pj** is faster throughout the interval from current date to previous switch implying the optimal execution would have switched rightaway, Optimal date vouches for this as it is same as the previous switch's date.

Switch 5 (DELAYED)**Original Switch:** 1992-03-13 → 1992-03-14

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-02-26)	11979.09 ms	12187.65 ms
q'_j (1992-02-27)	12066.00 ms	12041.30 ms

The optimal Date has been found in this interval **Pi** is faster at qi and **Pj** is faster at qj.

Switch 6 (EARLY)**Original Switch:** 1992-03-19 → 1992-03-20

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-05-19)	12235.22 ms	14009.16 ms
q'_j (1992-05-20)	12435.54 ms	13748.58 ms

Plan **Pi** is faster throughout the interval from the current date to next switch implying the optimal execution wouldn't perform any switches, Optimal date vouches for this as it is same as the next switch's date-1.

Switch 7 (DELAYED)**Original Switch:** 1992-05-20 → 1992-05-21

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-03-20)	15307.96 ms	12308.34 ms
q'_j (1992-03-21)	15067.45 ms	12108.91 ms

Plan **Pj** is faster throughout the interval from current date to previous switch implying the optimal execution would have switched rightaway, Optimal date vouches for this as it is same as the previous switch's date.

Switch 8 (DELAYED)**Original Switch:** 1992-12-04 → 1992-12-05

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1992-08-17)	12267.63 ms	12415.01 ms
q'_j (1992-08-18)	12285.29 ms	12320.98 ms

Plan **Pi** is faster throughout the interval from current date to the previous switch. which is impossible considering the plan cost is monotonous The only conclusion that can be made is there was a spike in **Pj**'s cost in this interval causing issue.

Switch 9 (DELAYED)

Original Switch: 1993-03-07 $\rightarrow$ 1993-03-08

Optimal Date	Plan P'_i	Plan P'_j
q'_i (1993-01-13)	12358.05 ms	12295.82 ms
q'_j (1993-01-14)	12392.70 ms	12372.49 ms

Plan **Pj** is faster throughout the interval from current date to previous switch implying the optimal execution would have switched rightaway, but in this case the optimal switch does not vouch for the previous switch and only justification I can give is that there must have been some date at which the cost is similar to the previous switch's cost.

Binary search is Performed instead of a linear one considering there are some parts of the data which are not monotonous is solely because of its humongous runtime.